Statistics Assignment –

Module 4: Essential Statistics & Mathematics for Data Science Theoretical Assignments:

1. Explain Key Statistical Concepts

→ 1. Types of data :

Qualitative data and Quantitative data .

Qualitative data means non -numerical data which is also known as categorical data eg. gender , colour.

Quantitative data means numerical data . It contains discrete(countable) eg. number of students and continuous (measurable) eg. height,weight.

→ 2. Measure of central tendency:

It tells the center of data.

It contains mean, median , mode.

Mean: sum/total values

Median : middle data after sorting.

Mode: Most frequent data.

→ 3. Measure of Dispersion:

It shows the spread of the data.

It contains variance,range and standard deviation.

Range means Max-min.

Variance means average squared difference of mean.

Standard deviation means squared root of variance.

→ 4. Probability:

Likelihood of an event(0 to 1).

→ 5. Population and sample:

Population means entire group.

Sample means subset of population.

→ 6. Hypothesis testing :

It is used to make decisions.

Null hypothesis (H0): No effect.

Alternative hypothesis(H1) : There is an event.

→ 7. Correlation :

Measures relationship between variables.

Values ranges from -1 to 1.

+1 strong positive .

-1 strong negative.

0 no relation.

→ 8. Regression :

Predicts one variable using another variable.

Eg. Predict marks based on study hours.

→ 9. Error in Statistics :

Type I error : Rejecting the true null hypothesis.

Type II error : Accepting the false null hypothesis.

→ 10. Confidence Interval :

Range in which the true value lies with certain confidence (EG. 95%).

2. Presentation: Importance of Probability in Machine Learning.

→ https://github.com/mansibhagwat006-ai/python-atmsystem/blob/main/Presentation%20-%20Importance%20of%20Probability%20in%20Machine%20Learning.pdf

Practical Tasks :

1. Car Rental Duration Analysis

Question: Find the mean, median, and mode of the rental durations.

Data contains n = 50 values and total sum = 172.

a) Mean

Formula: Mean = Sum of observations / Number of observations

Mean = 172 / 50 = 3.44 days

Answer: The average rental duration is 3.44 days.

b) Median

For 50 values, the median is the average of the 25th and 26th values in the sorted list.

25th value = 3, 26th value = 3

Median = (3 + 3) / 2 = 3 days

Answer: The typical rental duration is 3 days.

c) Mode

The mode is the value with the highest frequency.

Frequencies: 1 day = 4, 2 day = 13, 3 day = 11, 4 day = 9, 5 day = 7, 6 day = 4, 7 day = 2

Highest frequency = 13 for 2 days

Answer: The mode is 2 days, so 2-day rentals occur most often.

Conclusion: Most customers rent cars for about 2 to 3 days, so short-term rental plans may be the most useful for pricing and fleet planning.

2. Delivery Time Analysis

Question: Find the range, variance, and standard deviation of the shipment delivery times.

a) Range

Formula: Range = Maximum - Minimum = 7 - 1 = 6 days

Answer: The range of delivery times is 6 days.

b) Variance

Sample mean = 176 / 50 = 3.52

Sample variance formula: $s^2 = \text{Sum } (x - \bar{x})^2 / (n - 1)$

Here, $\text{Sum } (x - \bar{x})^2 = 114.48$

$s^2 = 114.48 / (50 - 1) = 2.34$

Answer: The sample variance of delivery times is 2.34 days^2.

c) Standard Deviation

Formula: Standard deviation = sqrt(variance)

s = sqrt(2.34) = 1.53

Answer: The sample standard deviation is 1.53 days.

Conclusion: Delivery times are spread around the mean by roughly one and a half days, which shows moderate variation in logistics performance.

3. Monthly Revenue Analysis

Question: Find the average monthly revenue and the range of revenue.

a) Measure of Central Tendency

Total revenue = 1590 thousand dollars

Average revenue = 1590 / 12 = 132.5 thousand dollars

Answer: The average monthly revenue is 132.5 thousand dollars.

b) Measure of Dispersion

Range = Maximum - Minimum = 155 - 110 = 45 thousand dollars

Answer: The range of monthly revenue is 45 thousand dollars.

Conclusion: The product earns about 132.5 thousand dollars per month on average, with a fluctuation of 45 thousand dollars between the highest and lowest months.

4. Fuel Efficiency of Vehicle Models

Question: Find the average, range, and variance of fuel efficiency for each vehicle model.

Model	Average mpg	Range	Sample variance
A	30.6	5	2.27
B	25.9	5	2.77
C	22.9	5	2.77
D	18.8	4	1.73
E	34.2	4	1.73

Model A

Mean = 306 / 10 = 30.6 mpg

Range = 33 - 28 = 5 mpg

Variance = 2.27

Model B

Mean = 259 / 10 = 25.9 mpg

Range = 28 - 23 = 5 mpg

Variance = 2.77

Model C

Mean = 229 / 10 = 22.9 mpg

Range = 25 - 20 = 5 mpg

Variance = 2.77

Model D

Mean = 188 / 10 = 18.8 mpg

Range = 21 - 17 = 4 mpg

Variance = 1.73

Model E

Mean = 342 / 10 = 34.2 mpg

Range = 36 - 32 = 4 mpg

Variance = 1.73

Conclusion: Model E has the highest average fuel efficiency, while Model D has the lowest. Models A to C show slightly higher variation than Models D and E.

5. Defect Frequency Analysis

Question: Create a bar chart and histogram, and identify the most common defect type.

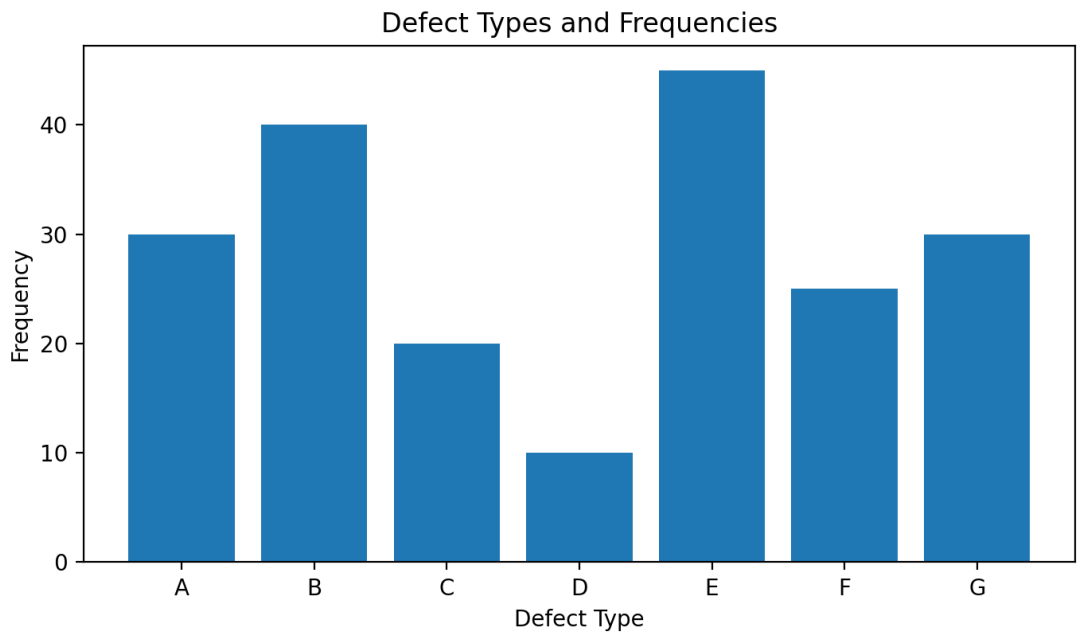
Frequency Table

Defect Type	Frequency

A	30
B	40
C	20
D	10
E	45
F	25
G	30

a) Bar Chart

The following bar chart shows the frequency of each defect type:



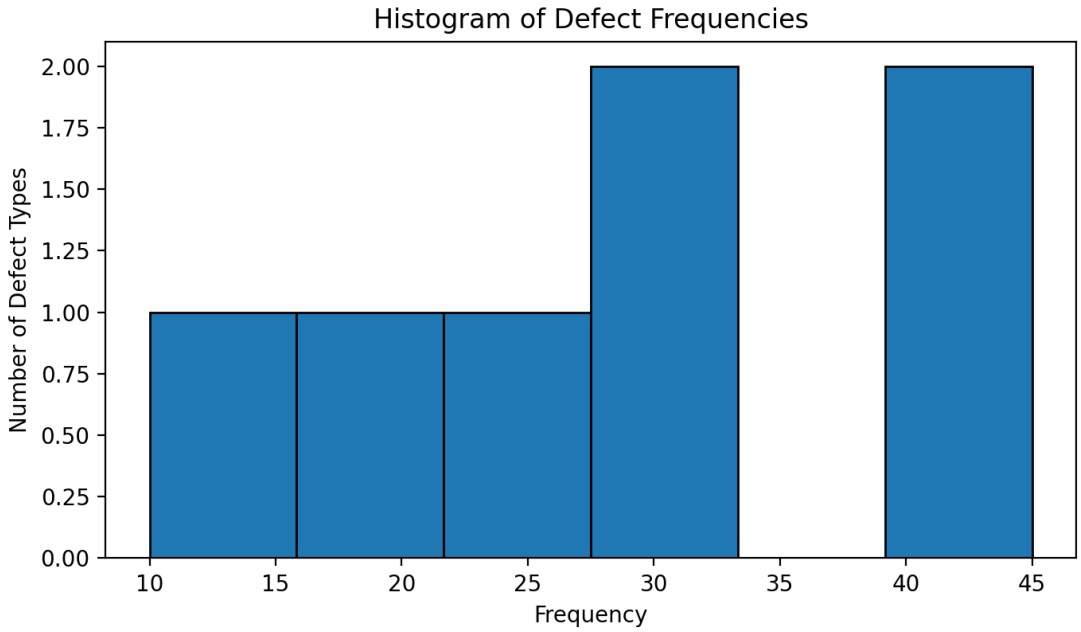
b) Most Common Defect

Highest frequency = 45, which corresponds to defect type E

Answer: Defect type E is the most common defect.

c) Histogram

The histogram below groups the defect frequencies to show how the frequencies are distributed:



Conclusion: Defect E appears most often, so it should be checked first during quality control.

6. Satisfaction Rating Analysis

Question: Calculate skewness and kurtosis of the customer satisfaction ratings and interpret the distribution.

a) Skewness

Skewness formula: g1 = m3 / (m2)^(3/2)

Using the rating data, skewness = -0.208

Answer: Skewness is approximately -0.208.

b) Kurtosis

Excess kurtosis formula: g2 = m4 / (m2)^2 - 3

Using the rating data, excess kurtosis = -0.768

Answer: Kurtosis is approximately -0.768.

c) Interpretation

The skewness is slightly negative, so the ratings lean a little toward the higher side. The kurtosis is negative, which means the distribution is flatter than a perfectly normal distribution. In simple words, most customers gave ratings around 3 to 5, and the responses are not extremely sharply peaked.

7. Weight Distribution Analysis

Question: Find Q1, Q2, Q3 and the 15th, 50th, and 85th percentiles of the weights.

a) Quartiles

Using position formula: Position = p/100 x (n + 1)

Q1 position = 25/100 x 101 = 25.25, so Q1 = 141.25 kg

Q2 position = 50/100 x 101 = 50.50, so Q2 = 267.50 kg

Q3 position = 75/100 x 101 = 75.75, so Q3 = 393.75 kg

Answer: Q1 = 141.25 kg, Median (Q2) = 267.50 kg, Q3 = 393.75 kg.

b) Percentiles

15th percentile position = 15/100 x 101 = 15.15, so P15 = 92.45 kg

50th percentile is the median, so P50 = 267.50 kg

85th percentile position = 85/100 x 101 = 85.85, so P85 = 444.25 kg

Answer: P15 = 92.45 kg, P50 = 267.50 kg, P85 = 444.25 kg.

c) Interpretation

About 25% of the individuals weigh less than Q1, 50% weigh less than the median, and 75% weigh less than Q3. The 85th percentile shows the upper part of the distribution, meaning only about 15% of the people have weights above that value.

8. Correlation Between Study Hours and Exam Scores

Question: Calculate the correlation coefficient and interpret it.

Calculation

Pearson correlation coefficient measures the strength and direction of a linear relationship.

Using the two data sets, correlation coefficient r = 0.9773

Answer: The correlation coefficient is 0.9773.

Interpretation: This is a very strong positive correlation. As study hours increase, exam scores also increase. So the relationship between studying and marks is strong and direct.

9. Probability of Getting at Least 8 Correct Answers

Question: A student guesses on a 10-question multiple-choice test with 4 options each. Find P(X >= 8).

Calculation

This is a binomial distribution problem.

n = 10, p = 1/4, q = 3/4

P(X >= 8) = P(X=8) + P(X=9) + P(X=10)

P(X=8) = 10C8 x (1/4)^8 x (3/4)^2 = 0.00038624

P(X=9) = 10C9 x (1/4)^9 x (3/4) = 0.00002861

P(X=10) = 10C10 x (1/4)^10 = 0.00000095

Total probability = 0.00038624 + 0.00002861 + 0.00000095 = 0.00041580

Answer: The probability of getting at least 8 correct answers is 0.00041580 (about 0.0416%).

10. Light Bulb Lifetime Probability

Question: Find the probability that a bulb lasts between 900 and 1100 hours when $X \sim N(1000, 100^2)$.

Calculation

Convert the values to z-scores:

$z1 = (900 - 1000) / 100 = -1$

$z2 = (1100 - 1000) / 100 = 1$

$P(900 < X < 1100) = P(-1 < Z < 1)$

From the standard normal table, $P(-1 < Z < 1) = 0.6827$

Answer: The probability is 0.6827, or about 68.27%.

11. Confidence Interval for Mean Height

Question: Calculate a 95% confidence interval for the population mean height.

Calculation

Formula: $\bar{x} \pm z \times (s / \text{sqrt}(n))$

Standard error = 8 / sqrt(100) = 0.80

Margin of error = 1.96 x 0.80 = 1.57

Confidence interval = 170 ± 1.57

Lower limit = 168.43, Upper limit = 171.57

Answer: The 95% confidence interval is (168.43 cm, 171.57 cm).

12. Hypothesis Test for Product Weight

Question: Test whether the sample provides evidence regarding the company's claim that the average weight is 500 grams.

Step 1: Hypotheses

$H0: \mu = 500$ grams

$H1: \mu \neq 500$ grams

Step 2: Test Statistic

$t = (\bar{x} - \mu) / (s / \text{sqrt}(n))$

$t = (510 - 500) / (20 / \text{sqrt}(25)) = 2.50$

Step 3: Decision

At 5% level of significance and df = 24, the two-tailed critical t value is about ±2.064.

Since calculated $t = 2.50 > 2.064$, reject H0.

Answer: There is significant evidence that the true average weight is different from 500 grams. The sample does not support the company's exact claim.

13. Hypotheses for a Teaching Method Study

Question: State the null and alternative hypotheses for checking whether the new teaching method improves student performance.

Hypotheses

Let μ_1 = mean score of students taught by the new method

Let μ_2 = mean score of students taught by the traditional method

H_0: $\mu_1 <= \mu_2$ (the new method does not improve performance)

H_1: $\mu_1 > \mu_2$ (the new method improves performance)

Answer: The null hypothesis says there is no improvement, while the alternative hypothesis says the new method gives better scores.

14. Confidence Interval for Population Proportion

Question: Calculate a 90% confidence interval for the proportion of people who support the policy.

Calculation

Formula: $\hat{p} \pm z \times \text{sqrt}[\hat{p}(1 - \hat{p}) / n]$

Sample proportion $\hat{p} = 320 / 500 = 0.64$

Standard error = sqrt[0.64 x 0.36 / 500] = 0.0215

Margin of error = 1.645 x 0.0215 = 0.0353

Confidence interval = 0.64 ± 0.0353

Lower limit = 0.6047, Upper limit = 0.6753

Answer: The 90% confidence interval is (0.6047, 0.6753).

15. Poisson Probability for Defects per Batch

Question: If defects per batch follow Poisson distribution with mean 2, find the probability of exactly 3 defects.

Calculation

Poisson formula: $P(X=x) = e^{(-\lambda)} \lambda^x / x!$

$P(X=3) = e^{(-2)} \times 2^3 / 3! = 0.180447$

Answer: The probability of getting exactly 3 defects is 0.180447 (about 18.04%).